

for some almost surely finite random variable O' .

Remark 6.8. Under the assumption that $\liminf_{i \rightarrow \infty} d(i) \geq R$ we can adapt the proof of (6.23) (and by using (6.1) rather than (6.2)) to show that for any $\varepsilon > 0$ there exists $p > 0$ such that

$$\lim_{t \rightarrow \infty} \mathbb{P}_{\mathcal{S}} \left(\forall s \in [(1-p)t, (1+p)t] : \left| \frac{O_s^{\text{cont}}}{t} - \frac{R}{\lambda^* + R} \right| < \varepsilon \right) = 1.$$

It yields a weaker bound on O_s^{cont} , but with a weaker condition on d and a larger range of s . ◀

Remark 6.9. When setting $s = t$ in (6.23) and (6.24), the quantity in the absolute values converges to zero in probability. Though in some cases where $\mathcal{K}$ grows sufficiently fast it may be possible to strengthen the convergence in probability to almost sure convergence, we are unable to derive the stronger bounds due to the use of (6.18) and (6.19) in Corollary 6.6. Stronger versions of these results would be required to improve to almost sure convergence. ◀

Remark 6.10. Proposition 6.7 provides the main step in proving the asymptotic behaviour of O_n , as presented in Theorems 2.3 through 2.10. It remains to translate the asymptotic behaviour of O_t^{cont} into the asymptotic behaviour of O_n , which is carried out in Section 9. ◀

Intuitively, an individual born at time $T < t$ survives up to time t with probability $\mathbb{P}(L > t - T)$, independently of all other individuals. At the same time, the number of individuals born around time T is roughly $e^{\lambda^* T}$ by Proposition 6.5. The proof uses first and second moment bounds to establish the optimal choice of T such that no individuals born before T are alive at time t , whereas many individuals born after T are alive at time t .

Proof. We first prove (6.23). Fix $K, p > 0$ and define

$$\ell_t := \frac{R}{\lambda^* + R}t - K \log t, \quad u_t := \frac{R}{\lambda^* + R}t + K \log t, \quad \tilde{u}_t := \frac{R}{\lambda^* + R}t, \quad (6.26)$$

and

$$\underline{r}(t) := t - p \log t, \quad \bar{r}(t) := t + p \log t.$$

We then observe that O_t^{cont} is increasing in t , so that

$$\mathbb{P}_{\mathcal{S}} \left(\left| O_s^{\text{cont}} - \frac{R}{\lambda^* + R}t \right| < K \log t \text{ for all } s \in [\underline{r}(t), \bar{r}(t)] \right) \geq 1 - \mathbb{P}_{\mathcal{S}} \left(O_{\underline{r}(t)}^{\text{cont}} \leq \ell_t \right) - \mathbb{P}_{\mathcal{S}} \left(O_{\bar{r}(t)}^{\text{cont}} \geq u_t \right). \quad (6.27)$$

We thus aim to bound the probabilities on the right-hand side from above, starting with the leftmost one. For any $\delta > 0$, recalling that $\mathcal{A}_t^{\text{cont}}$ denotes the set of alive individuals at time t ,

$$\begin{aligned} \mathbb{P}_{\mathcal{S}} \left(O_{\underline{r}(t)}^{\text{cont}} \leq \ell_t \right) &= \mathbb{P}_{\mathcal{S}} \left(\mathcal{B}(0, \ell_t) \cap \mathcal{A}_{\underline{r}(t)} \neq \emptyset \right) \\ &\leq \frac{1}{\mathbb{P}(\mathcal{S})} \left[\mathbb{P} \left(\mathcal{B}(0, \ell_t) \cap \mathcal{A}_{\underline{r}(t)} \neq \emptyset, |\mathcal{B}(0, \ell_t)| \leq e^{\lambda^* \ell_t + \delta \log t} \right) \right. \\ &\quad \left. + \mathbb{P} \left(|\mathcal{B}(0, \ell_t)| \geq e^{\lambda^* \ell_t + \delta \log t} \right) \right]. \end{aligned} \quad (6.28)$$

The second probability in the brackets converges to zero by Corollary 6.6 (with $r = 0, s = \ell_t$, and $u = \delta \log t$). Furthermore, lifetimes among individuals are i.i.d. and each individual born before time ℓ_t needs to live for at least $\underline{r}(t) - \ell_t$ time to be alive at time $\underline{r}(t)$. Hence, by a union bound and using the upper bound in (6.2) of Lemma 6.1, we arrive for some large constant $C > 0$ at the upper bound

$$\mathbb{P}(\mathcal{S})^{-1} e^{\lambda^* \ell_t + \delta \log t} \mathbb{P}(L \geq \underline{r}(t) - \ell_t) + o(1) \leq e^{\lambda^* \ell_t + \delta \log t - R(\underline{r}(t) - \ell_t) + C \log(\underline{r}(t) - \ell_t)} + o(1). \quad (6.29)$$

By the definition of ℓ_t and $\underline{r}(t)$, this equals

$$\exp \left(\left[-(\lambda^* + R)K + \delta + Rp + C \right] \log t + \mathcal{O}(1) \right) + o(1).$$

Hence, for any K larger than $K_0 := C(\lambda^* + R)^{-1}$, we can choose δ and p sufficiently small so that the terms in the square brackets are negative. As a result, the upper bound in (6.29) tends to zero with t for these choices of K, p , and δ .